

# Backtracking, Modular Arithmetic, Multiplicative Inverse

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# Overview

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- 2 Fundamental Theorem of Arithmetic (8.4)
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- 4 Arithmetic with a Prime Modulus (8.6)
  - Multiplicative Inverses (8.6.1)

## GCD Linear Combination Theorem

A refresher from the end of last class:

**Theorem:** The greatest common divisor of  $a$  and  $b$  is a linear combination of  $a$  and  $b$ .  
That is,  $\gcd(a, b) = s \cdot a + t \cdot b$  for some integers  $s$  and  $t$ .

## Computing the key

We can use this theorem as an algorithm to find the linear combination of  $a$  and  $b$  that produces their GCD.

```
def gcdcombo( $a, b$ ):  
    if  $a = 0$ : return( $0, 1, b$ )  
    else:  
        ( $s, t, g$ ) = gcdcombo( $\text{rem}(b, a), a$ )  
        return( $t - s \cdot \text{qcnt}(b, a), s, g$ )
```

- $\text{gcdcombo}(0, 15) = (0, 1, 15)$
- $\text{gcdcombo}(10, 15) = (-1, 1, 5)$
- $\text{gcdcombo}(24, 64) = (3, -1, 8)$

## Computing By Hand

| $a$ | $b$ | $s$ | $t$ | $g$ |
|-----|-----|-----|-----|-----|
| 24  | 64  | 3   | -1  | 8   |
| 16  | 24  | -1  | 1   | 8   |
| 8   | 16  | 1   | 0   | 8   |
| 0   | 8   | 0   | 1   | 8   |

Do the rems going down, then the qcnts going up. Note that, at every level:  $sa + tb = g$  (sanity check!).

## Pulvarizing

**Corollary:** An integer is a linear combination of  $a$  and  $b$  iff it is a multiple of  $\gcd(a, b)$ .

**Proof:**

Let  $g = \gcd(a, b)$ . We showed  $g = sa + tb$  for some  $s$  and  $t$ . Any multiple of  $g$  is a linear combination of  $a$  and  $b$ :  $kg = k(sa + tb) = (ks)a + (kt)b$ .

We know  $a = k_1g$  and  $b = k_2g$  because  $g$  is a common divisor of  $a$  and  $b$ . Any linear combination of  $a$  and  $b$  is a multiple of  $g$ :  $s'a + t'b = s'(k_1g) + t'(k_2g) = (s'k_1 + t'k_2)g$ .

Mixing  $a$  and  $b$  in different combinations, we can only make multiples of  $g$ .

Note: The combinations are not unique:  $sa + tb = (s - b)a + (t + a)b$ .

# Fundamental Theorem of Arithmetic

**Theorem:** Every integer greater than 1 is a product of a unique non-increasing sequence of primes.

**Lemma:** If  $p$  is a prime and  $p|ab$ , then  $p|a$  or  $p|b$ .

**Proof of Lemma:** One case is if  $\gcd(a, p) = p$ . Then, the claim holds, because  $a$  is a multiple of  $p$ .

Otherwise,  $\gcd(a, p) \neq p$ . In this case,  $\gcd(a, p)$  must be 1, since 1 and  $p$  are the only positive divisors of  $p$ . Since  $\gcd(a, p)$  is a linear combination of  $a$  and  $p$ , we have  $1 = sa + tp$  for some  $s, t$ . Then,  $b = s(ab) + (tp)b$ ; that is,  $b$  is a linear combination of  $ab$  and  $p$ . Since  $p$  divides both  $ab$  and  $p$ , it also divides their linear combination,  $b$ . QED.

## Proof of Fundamental Theorem of Arithmetic

**Lemma:** Let  $p$  be a prime. If  $p|a_1a_2 \cdots a_n$ , then  $p$  divides some  $a_i$ .

**Proof:** Every positive integer can be expressed as a product of primes. (Challenge: strong induction!) We need to show this expression is unique. We proceed by contradiction: Assume there exists positive integers that can be written as products of primes in more than one way. Take the smallest such integer  $n$  and let  $n = p_1p_2 \cdots p_j = q_1q_2 \cdots q_k$  be the two decompositions. Arrange them in non-increasing order and assume without loss of generality that  $p_1 \leq q_1$ . If  $p_1 = q_1$ , the remaining part of the product is smaller than  $n$  and different, which is a contradiction ( $n$  was the smallest).

Note that all the  $p_i$ s are less than  $q_1$ . But  $q_1|n$  and  $n = p_1p_2 \cdots p_j$ , so  $q_1$  divides one of the  $p_i$ s, which contradicts the fact that  $q_1$  is bigger than all them. QED.



# Congruence definition

Definition:  $a$  is *congruent to  $b$  modulo  $n$*  iff  $\text{rem}(b, n) = \text{rem}(a, n)$ . Equivalently,  $n \mid (a - b)$ .

We write  $a \equiv b \pmod{n}$ .

$29 \equiv 15 \pmod{7}$  because  $7 \mid (29 - 15)$ , namely 14. Both have a remainder of 1 when divided by 7.

Equivalence relation—partitions the integers.

Transitivity, reflexivity, symmetry.

## Basic modular algebra

In regular algebra,

$$a = b$$

$$a + c = b + c.$$

Can we do the same in congruence-land?

$$\begin{aligned} a &\equiv b \pmod{n} \\ a + c &\equiv b + c \pmod{n}. \end{aligned}$$

Yes!

$$\begin{aligned} a \equiv b \pmod{n} &\text{ iff } n \mid (a - b) \text{ iff } \exists k, kn = a - b \text{ iff } \exists k, kn = a - b + (c - c) \text{ iff} \\ &\exists k, kn = (a + c) - (b + c) \text{ iff } n \mid ((a + c) - (b + c)) \text{ iff } a + c \equiv b + c \pmod{n}. \end{aligned}$$

Multiplication is repeated addition, so we can also multiply both sides by a constant. By transitivity, we can always add or multiply each side by values that are congruent!  
“Clock arithmetic”.

## Example

$$2x + 17 \equiv x + 31 \pmod{12}$$

$$2x \equiv x + 14 \pmod{12} \quad \text{add } -17 \text{ to both sides}$$

$$2x \equiv x + 2 \pmod{12} \quad \text{add } 0 \text{ to left and } -12 \text{ to right}$$

$$x \equiv 2 \pmod{12} \quad \text{add } -x \text{ to both sides}$$

Double check.  $4 + 17 = 21$  vs.  $33$ . Difference is  $12$ , check!

$$3x + 4 \equiv 27 \pmod{11}$$

$$3x \equiv 23 \pmod{11} \quad \text{add } -4 \text{ to both sides}$$

Kind of stuck because we don't (yet) have a "divide both sides by 3" rule.

## So, what about division?

If  $a \equiv b \pmod{n}$ , can we divide both sides by  $c$ ?

$$7 \equiv 28 \pmod{3}$$

$$1 \equiv 4 \pmod{3} \quad \text{divide by 7}$$

So, maybe? At least if the answers are integers?

Is division even meaningful more generally?

## Back to basics

Definition: The *multiplicative inverse* of a number  $x$  is a number  $x^{-1}$  such that:  
 $x \cdot x^{-1} = 1$ .

Division by  $x$  is really multiplication by  $x^{-1}$ .

Over the reals, what values have inverses? Everybody but zero.

Over the integers, what values have inverses? Only 1 and  $-1$ .

Over the integers mod  $n$ , what values have inverses?

## Example, mod 10

|   | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 2 | 0 | 2 | 4 | 6 | 8 | 0 | 2 | 4 | 6 | 8 |
| 3 | 0 | 3 | 6 | 9 | 2 | 5 | 8 | 1 | 4 | 7 |
| 4 | 0 | 4 | 8 | 2 | 6 | 0 | 4 | 8 | 2 | 6 |
| 5 | 0 | 5 | 0 | 5 | 0 | 5 | 0 | 5 | 0 | 5 |
| 6 | 0 | 6 | 2 | 8 | 4 | 0 | 6 | 2 | 8 | 4 |
| 7 | 0 | 7 | 4 | 1 | 8 | 5 | 2 | 9 | 6 | 3 |
| 8 | 0 | 8 | 6 | 4 | 2 | 0 | 8 | 6 | 4 | 2 |
| 9 | 0 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |

What specific values have inverses? 1, 3, 7, 9.

What specific values do *not* have inverses? 0, 2, 4, 5, 6, 8.

General rule?  $a$  has an inverse (mod  $n$ ) iff  $\gcd(a, n) = 1$ .

## Back to solving

$$3x + 4 \equiv 27 \pmod{11}$$

$$3x \equiv 23 \pmod{11} \quad \text{add } -4 \text{ to both sides}$$

Want to multiply both sides by  $3^{-1} = 4$ , since  $3 \times 4 \equiv 1 \pmod{11}$ .

$$3x \equiv 23 \pmod{11}$$

$$4 \times 3x \equiv 4 \times 23 \pmod{11} \quad \text{multiply both sides by 4}$$

$$12x \equiv 92 \pmod{11} \quad \text{simplify}$$

$$x \equiv 4 \pmod{11} \quad \text{congruence}$$

Double check: 16 vs. 5, 11 divides the difference!